Extra Dimensions in Space and Time

Multiversal Journeys

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Extra Dimensions in Space and Time

Foreword by Lawrence M. Krauss

Springer

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To future scientists: Connor & Nima.

Farzad

To my family:

Itzhak

To: Laura, Jackson, and Maya.

John

Foreword

It is a pleasure to write this foreword for Multiversal Journeys' first book to further its goal of educating the public on the latest in theoretical physics results. I was privileged to participate in the lecture series on which this book is based, and enjoyed working, on separate occasions, along with John Terning and Itzhak Bars to carry on afternoon-long sessions with interested members of the public on issues ranging from extra dimensions to the future of the universe.

The lecture series was unique in that lengthy question and answer sessions were added to the lectures, to give the audience a chance to truly explore these fascinating areas in depth. The Multiversal book series is equally unique, providing book-length extensions of the lectures with enough additional depth for those who truly want to explore these fields, while also providing the kind of clarity that is appropriate for interested lay people to grasp the general principles involved.

As I have described in my own popular book, *Hiding in the Mirror*, and as both Itzhak Bars and John Terning colorfully describe in their reviews of the physics motivations for considering extra dimensions in this volume, humans have been fascinated for centuries by the possibility that there are extra dimensions, beyond the three spatial dimensions and one time dimension of our experience.

Over the past 90 years, beginning with the seminal work of Theodore Kaluza and Oskar Klein, this possibility has found its way into the heart of physics. The realization that gravity was associated with the curvature of space suggested that perhaps the other forces in nature might be associated with curvature in other, unobservable dimensions.

Starting in the 1960s an effort to deal with nasty infinities in quantum field theories pointed toward the possibility that fundamental elementary particles are in fact excitations of more fundamental units, string-like objects whose quantum vibrations would determine the spectrum of observed particles. It turned out that quantum strings, however, only make mathematical sense and only remove infinities if the number of space–time dimensions is much greater than four. In the original string models one required 26 dimensions!

Once supersymmetry was discovered as a possible symmetry of nature, it became possible to reduce the critical number of dimensions for viable strings to 10 or 11 dimensions. Since that time it has become clear that dimensionality itself may be illusory, and that what the physics encoded in a given number of dimensions may in fact be completely described by a projection onto one less dimension, just as a hologram encodes all the 3D information associated with an object on a 2D plate.

Of course, once the possibility of other space–time dimensions was proposed, the obvious question then became, where are they and how come we cannot detect them? Klein's first answer formed the basis of most of the subsequent work in this area. If extra dimensions of space are "curled up" on scales smaller than we can detect with our probes then they could effectively remain invisible.

More recently another more dramatic possibility was proposed. Gravity is by far the weakest force known in nature. If somehow all the other forces were restricted to our 4D manifold, and only gravity could propagate in extra dimensions then once again these dimensions could have been thus far undetected. Moreover, the size of the other dimensions would no longer be restricted to be small. They could be infinitely big!

In addition, the fact that gravity could propagate in more than three spatial dimensions also suggested a possible explanation for why gravity is so weak in our universe. Instead of a $1/r^2$ falloff that we see on scales in which we have thus far measured gravity, if gravity propagated in higher dimensions on smaller scales, it would experience a faster falloff. Thus, in fundamental scales the strength of gravity could be comparable to that of the other forces in nature and yet on scales we can measure it today it would appear far weaker.

One final exotic possibility remains. Highest dimensional theories involve more than three spatial dimensions but one time dimension. Some people have wondered whether it might be possible, in fact, to derive sensible laws of physics in a universe with higher dimensions and two or more "time-like" dimensions. This remained as an interesting but controversial possibility for a long time, but Itzhak Bars has solved these problems in his formulation of Two Time Physics with a new powerful gauge symmetry in phase space. He has been able to construct a physically meaningful framework that includes one extra time and one extra space dimension. As explained by Bars in this volume, successful physical theories in 3+1 dimensions now have counterparts in 4+2 dimensions. These theories make interesting predictions for observable physical phenomena that may extend our understanding of physics in three spatial dimensions and one time dimension.

Of course, it is necessary to add here that in spite of the remarkable theoretical interest in extra dimensions, and their great aesthetic appeal to many scientists and many members of the general public, no *direct* experimental evidence whatsoever yet exists for their presence (although Bars has argued that his Two Time Physics actually predicts observed relations in our world). Thus the question of whether they are remarkably interesting mathematical possibilities or whether they reflect an important underlying fundamental physical reality is currently not universally agreed upon by physicists. Of course, this is the price we pay when we explore the very limits of our understanding of nature.

Tempe, Arizona, USA Lawrence M. Krauss
July 2009

Preface

There is a hidden sun in a particle
Suddenly the particle opens up its mouth

The earth and the heavens are cut into pieces
In presence of the sun escaping its trap

Rumi
1207–1273

One of the most popular topics in Multiversal Journey's lecture series has been the concept of extra dimensions in space and time. The topic has been covered in many of our conferences within the last few years: four times by the authors of this book.

The word dimension comes from the 14th century Latin *dimetiri* meaning to measure out. Historically, the notion of a dimension has long been used in geometry for centuries. The dimension of an object is specified by its coordinates (degrees of freedom: longitude, latitude, and height). In algebra, the notion of a dimension takes an abstract form such as dimensions of a vector space which are not the same dimensions we experience everyday in life.

In physics, the idea of extra spatial dimensions originates from Nordstöm's 5-dimensional vector theory in 1914, followed by Kaluza–Klein theory in 1921, in an effort to unify general relativity and electromagnetism in a 5-dimensional space–time (4 dimensions for space and 1 for time). The Kaluza–Klein theory didn't generate enough interest with physicist for the next five decades, due to its problems with inconsistencies. With the advent of supergravity theory (the theory that unifies general relativity and supersymmetry theories) in late 1970s and eventually, string theories (1980s), and M-theory (1990s), the dimensions of space–time increased to 11 (10-space and 1-time dimension).

In contrast, to the volume of research that has been conducted in the area of extra spatial dimensions in the last 40 years, not much time has been devoted to multi-dimensional time theories. Earlier attempts in multi-dimensional time theories had problems with violating causality among other issues. Two-time physics, a theory with 4-space and 2-time dimensions, is well covered in this book.

There are two main features in this book that differentiates it from other books written about extra dimensions:

The first feature is the coverage of extra dimensions in time (two-time physics), which has not been covered in earlier books about extra dimensions. All other books mainly cover extra spatial dimensions.

The second feature deals with the level of presentation. The material is presented in a non-technical language followed by additional sections (in the form of appendices or footnotes) that explain the basic equations and principles. This feature is very attractive to readers who want to find out more about the theories involved beyond the basic description for a layperson. The text is designed for scientifically literate non-specialists who want to know the latest discoveries in theoretical physics in a non-technical language. Readers with basic undergraduate background in modern physics and quantum mechanics can easily understand the technical sections.

The two parts of the book can be read independently. One can skip Part I and go directly to Part II which covers extra dimensions in space.

Part I starts with an overview of the standard model of particles and forces, notions of Einstein's special and general relativity, and the overall view of the universe from the Big Bang to the present epoch (Chapters 1, 2 and 3).

Chapter 4 covers basics of symmetry and perspective including local, global, and gauge symmetries.

Beyond the first four chapters, Part I covers two-time physics (Chapters 5, 6, 7, 8, 9, and 10). Two-time physics is the heart of Part I of this book and is best described by its author, Prof. Bars:

> Humans normally perceive physical reality in 3 space and 1 time dimensions and this is encoded in equations of physics in 3+1 dimensions (1T-physics). However, as discussed in this book, 1T-physics systematically misses to predict certain additional real phenomena in 3+1 dimensions in the form of hidden symmetries and hidden relations between apparently different dynamical systems.
>
> 2T-physics, which is based on a fundamental symmetry that can be realized only by adding an extra space and an extra time dimensions, is a completion of 1Tphysics that captures the missing information as effects due to the extra dimensions.
>
> According to 2T-physics in 4+2 dimensions, using 1T-physics in 3+1 dimensions is like analyzing only "shadows" on walls by observers stuck on "walls=3+1", while using 2T-physics is like analyzing directly the "substance" in the room by observers in the "room=4+2". The more powerful perspective of being in the "room" makes predictions, beyond those of 1T-physics, that can be tested and verified directly in our own 3+1 dimensional spacetime.
>
> With proper interpretation, some observations in our own spacetime can be used to peek into and explore the extra 1+1 dimensions which are neither small nor hidden.
>
> 2T-physics has worked correctly at all scales of physics, both macroscopic and microscopic, for which there is experimental data so far. In addition to revealing hidden information even in familiar "everyday" physics, it also makes testable predictions in lesser known physics regimes that could be analyzed at the energy scales of the Large Hadron Collider at CERN or in cosmological observations.

The technical sections in Part I are provided in the form of footnotes.

Part II of the book is focused on extra dimensions of space. As was mentioned earlier, one can skip Part I and go directly to Part II. It covers the following topics:

- The Popular View of Extra Dimensions
- Einstein and the Fourth Dimension
- Traditional Extra Dimensions
- Einstein's Gravity
- The Theory Formerly Known as String
- Warped Extra Dimensions
- How Do We Look for Extra Dimensions?

The technical section for Part II is covered at the end under, extra material: the equations behind the words.

I am indebted to Professor Lawrence M. Krauss for writing the foreword to the book. I would like to extend my thanks to the other two members of the advisory council for the book: Professor Mark Trodden and Professor David Finkelstein.

I would also like to thank the staff of Springer, especially Jeanine Burke and Dr. Harry Blom for making this project happen.

I am grateful to the following people associated with Multiversal Journeys who helped this project immensely:

- Multiversal Journeys, board of directors, professor Gene Moriarty, Richard D. Holt, and Dr. Faranak Nekoogar for careful review of the manuscript.
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- Kai Barzilai for his technical support.

Finally, I would like to thank my father for providing the quote from Rumi and his encouragement for this project.

Multiversal Journeys Farzad Nekoogar
Encino, California, USA
July 2009

Contents

Part I
Two-Time Physics: The Unified View from Higher Dimensional Space and Time

Itzhak Bars

Foreword to 2T-Physics

This part of the book aims at making the notions of 2T-physics more accessible to interested people. The exposition is mostly descriptive, but there will be some formulas (mostly in footnotes) to satisfy the curiosity of those who can follow them. The book is written at a pedagogical level that non-experts with little or no background in physics should be able to follow and enjoy for the most part. There are, however, some unfamiliar notions that may require more concentration when some of the more technical topics are discussed. Those topics will repeatedly come up, so it is likely that the ideas will sink in gradually as the reader makes more progress. Of course, readers with some prior knowledge on space–time, extra dimensions, string theory, or particle physics will be able to follow all aspects of the theory more easily. Physicists from every field as well as scientists or engineers with some mathematical background, who can understand the formulas, are very likely to get a considerable deeper insight on the ideas discussed in this book.

In this foreword I will first describe the notions of 2T-physics in a nutshell and then provide a road map to the various chapters.

Humans normally perceive physical reality as 4 dimensional, namely 3-dimensional space: up–down, back-forth, and side-to-side, plus 1-dimensional time: past–future. It is therefore natural that the ordinary formulation of physics, which I will call 1-time physics (1T-physics), has been developed in 3-space and 1-time dimensions. However, evidence has been accumulating that 3+1 dimensions are insufficient to describe our world, just like shadows on walls alone are insufficient to capture the true essence of an object in a 3-dimensional room.

Two-time physics (2T-physics) is compatible with 1T-physics, but it proposes a more fundamental formulation of physics in a 6-dimensional universe, comprised of 4-dimensional space and 2-dimensional time. Phenomena in 4+2 dimensions appear to observers in 3+1 dimensions the same way as an object in a room would appear as its own shadows on the surrounding walls. Naturally each distorted shadow, taken one at a time, contains considerably less data about its source. Similarly, 1T-physics in 3+1 dimensions describes each "shadow" one at a time and appears unable to capture all the information coming from 4+2 dimensions. Indeed there is more information in the real world in 3+1 dimensions, in the form of hidden symmetries and "dualities," that are missed systematically in 1T-physics. The hidden symmetries are distorted forms of higher dimensional symmetries and additional properties of higher dimensions inherent in each "shadow," while dualities are hard to notice relations among the "shadows." These additional phenomena are predicted by 2T-physics, while 1T-physics is only able to verify them but unable to predict them. The more encompassing and unifying 2T-physics is a completion of 1T-physics.

According to 2T-physics, using 1T-physics is like analyzing only "shadows on walls" by observers stuck on "walls," while using 2T-physics is like analyzing directly the "substance in the room" by observers in the "room." So, describing our common experiences by formulating them in the form of 1T-physics is like relying on restricted observers that are stuck in 3+1 dimensions and viewing a highly symmetric 4+2-dimensional system from a less favorable non-symmetric 3+1 perspective.

More generally, 2T-physics in d+2 dimensions is a completion of 1T-physics in (d–1) space and 1-time dimensions. The extra (d–1) – 3 space dimensions beyond the familiar 3 dimensions can be interpreted similar to the hidden dimensions of string theory. However, the additional extra space and extra time dimensions that takes us from 1T-physics to 2T-physics are neither small nor hidden; they are visible indirectly by proper interpretation of observations in our own 3+1-dimensional space–time.

While agreeing with the ordinary formulation of physics in 3+1 dimensions, 2T-physics reveals the hidden information as mentioned above. Some of those are easy to verify in macroscopic and microscopic physics as will be discussed in this book. In addition, 2T-physics also makes subtle predictions that make it distinguishable from other approaches in the description of lesser known regimes of physics. Therefore 2T-physics could also be tested through both computations and experiments at the level of particle physics at the energy scales of the Large Hadron Collider as well as in cosmology.

2T-physics emerges from requiring a new fundamental symmetry in the equations of motion in classical or quantum mechanics. This is a symmetry between covariant position (X^M = time & space) and contravariant momentum (P_M = energy & momentum). Such a symmetry is hinted in parts of 1T-physics, as in canonical transformations in the commutation rules that define quantum mechanics. Requiring that an appropriate extension of this symmetry must be obeyed at *every instant* in all equations that describe *all motion* is the most important new ingredient that uniquely leads to the equations of 2T-physics. The phenomena missed systematically by 1T-physics in 3+1 dimensions are captured by the requirement of this new *gauge symmetry* that acts on the phase space degrees of freedom (X^M,P_M) in 4+2 dimensions. The prediction of an extra time and an extra space dimensions as compared to those in 1T-physics is a direct consequence of this symmetry.

The 2T-physics approach provides new mathematical tools and new insights for understanding our universe at all scales of distance or energy. Evidence of the

4+2-dimensional world can be found at both the macroscopic and microscopic scales in the form of hidden symmetries and "dualities." Examples include the origin of symmetries that control the behavior of planets in celestial mechanics, the behavior of electrons in the hydrogen atom, and the behavior of quarks, leptons, and force particles in particle physics and cosmology. These are explained as hidden symmetries in 3+1 dimensions inherited from obvious symmetries in 4+2 dimensions, thus providing partial evidence for the existence of higher space–time.

Field theory is the most successful language so far for discussing the fundamental laws of nature, including the known four forces and the known fundamental matter in the form of quarks, leptons, and force particles. Applying the consequences of the new symmetry in particle mechanics to field theory has led to 2T-field theory, as a structure above 1T-field theory, with the power of making new predictions that are missed systematically in 1T-field theory. Amazingly, the best understood fundamental theories in physics that correctly describe nature as we know it today in 3+1 dimensions, namely the standard model of particles and forces and general relativity, have been shown to be "shadows" of more symmetric field theories in 4+2 dimensions. So, 2T-physics has worked correctly at all scales of physics for which there is experimental data so far.

The following is a road map to the various chapters.

To set in context the issues addressed and problems solved by 2T-physics, I have included some introductory material on past and current understanding of our universe from the very small to the very large. This includes a description of some of the historical evolution of notions on fundamental laws and principles, the current experimentally established structure of matter up to quarks, leptons, and force particles, the standard model of particles and forces, notions of Einstein's special and general relativity, and the overall view of the universe from the Big Bang to the present epoch. This material appears in Chapters 1, 2, and 3.

Since symmetry in physics and perspectives of observers are of fundamental importance to convey the notions of 2T-physics, I have included an elementary discussion of "global" symmetry and "gauge" symmetry in Chapter 4. It contains some discussion of how gauge symmetry appears in various theories including general relativity, the standard model, string theory, and 2T-physics. Gauge symmetry is the fundamental principle that leads to the existence and the detailed description of all the known forces and interactions. It is also the reason for the appearance of extra dimensions in string theory and in 2T-physics, as discussed in Chapters 5 and 6.

The shadows allegory is a helpful device to explain to the non-expert the relation between 2T-physics and 1T-physics. This allegory is discussed in Section 5.2 and is referred to in many later discussions in the rest of the book.

Although 2T-physics is already partially discussed in the earlier chapters, a systematic and considerably more detailed exposition is given in Chapter 7. This includes an informal explanation based on the shadows allegory and a formal discussion that displays some formulas. More details on the formulas and some related material appear in the footnotes. Consequences of 2T-physics, the missed information in 1T-physics, and the unifying aspects of 2T-physics are illustrated with one specific example that leads to Fig. 7.7. This figure captures concisely the main new notions of 2T-physics that apply not only to the specific example but also to all physical circumstances. In this way we see that 2T-physics is indeed a unifying structure that stands above 1T-physics.

In Chapter 8, some concrete examples of how to "see" 4+2 dimensions in everyday physics in 3+1 dimensions are discussed. Indeed, for the hydrogen atom and the celestial mechanics of planets, I show that observers in 3+1 dimensions can easily interpret their experimental observations as revealing information about the higher space–time in 4+2 dimensions. Other cases that appear in Fig. 7.7 are not as easy to describe with words or drawings, but with little mathematical effort and the guidance provided by 2T-physics, it is straightforward to learn about the presence of an extra time and an extra space dimension from observations directly in 3+1 dimensions plus the proper interpretation. These illustrate how we can *indirectly* see the effects of 4+2 dimensions.

Chapter 9 shows how to relate 2T-field theory to 1T-field theory through the shadow analogy, and in particular how to describe the standard model and general relativity in 2T-physics directly in 4+2 dimensions. The resulting shadow 1T field theories in 3+1 dimensions successfully describe all known aspects of fundamental physics as we know it today. But there are some new restrictions on scalar fields, such as the Higgs field and others that participate in cosmic phase transitions in the history of the universe. These new restrictions may help distinguish 2T-field theory from the usual less restrictive 1T-field theory through tests that may be conducted at the Large Hadron Collider at CERN or in cosmological observations.

Finally Chapter 10 covers recent progress as well as conceptual and philosophical directions to be pursued in 2T-physics. Progress in the past couple of years has extended from the 2T standard model and 2T-gravity to grand unification, supersymmetry, including higher dimensional super Yang–Mills theory up to 10+2 dimensions. The latter cannot be addressed by 1T-physics at all. Remaining challenges, the completion of the 2T-physics formalism for supergravity, strings, and branes, are also discussed. Future directions, which include the development of new computational techniques based on the hidden dualities and hidden symmetries, as well as conceptual and philosophical notions, are outlined.

For further reading, a complete list of my original papers on 2T-physics is included at the end of the book. The reader interested in more details should be able to find the desired information in those papers. Furthermore an index that points to the occurrence in the text of many physics or mathematics terms that may be unfamiliar to the non-expert should be helpful for both educational and informative purposes.

2T-physics sprouted in 1995 from my attempts to understand more deeply the hints for 2 time dimensions in the extended supersymmetry of M-theory. These efforts led in 1996 to my proposal of S-theory which was an algebraic approach to symmetries with two times unifying some duality aspects of M-theory. The attempt to find dynamics that overcomes fundamental problems with two times is what led to the new gauge symmetry. I would like to thank my past collaborator Kostas Kounnas who in 1997 at CERN participated in the initial stages of research for a gauge

symmetry. That effort eventually led in 1998 to the gauge symmetry now used in 2T-physics. My former student Cemsinan Deliduman, postdocs Oleg Andreev and Djordge Minic, and colleague Soo-Jong Rey, who collaborated with me on a number of papers at the early stages of 2T-physics developing crucial notions, deserve special mention. More recent progress, including the all important 2T standard model and 2T-gravity, was sparked by numerous productive discussions with my recent students and collaborators Shih Hung Chen, Yueh Cheng Kuo, Bora Orcal, Moises Picón, and Guillaume Quélin, to whom I am deeply indebted.

I should also mention influential discussions on many occasions over the years with Edward Witten and John Schwarz who have encouraged me to pursue my 2T-physics ideas, and conversations with Joe Polchinski, David Gross, as well as my USC colleagues Nicholas Warner, Clifford Johnson, and Krzysztof Pilch who have challenged me with questions that led to better and firmer results.

Finally, many thanks to colleagues, friends, students, and family members who read the manuscript and made recommendations that improved this book. In this regard I would like to thank Professor Werner Däppen and Dr. Douglas Burke who provided the insights of physicists from another field, my students mentioned in the previous paragraph who corrected many mistakes, and especially my son James Barsimantov and my wife Annie Rosenschein-Bars who represented the perspective of non-experts and provided suggestions for significant improvements.

Los Anseles, California, USA Itzhak Bars
June 14, 2009

Chapter 1
A First Look at the Known Universe

Among the foundational questions in physics and cosmology today is the deep mystery of why we live in a space–time of 3 space dimensions and 1 time dimension? Nowadays physicists ask seriously whether there are more space or time dimensions, and go further to suggest ways in which extra dimensions could play a role in determining the properties of the universe we observe today.

What leads us to even ask such questions and adopt such views?

For thousands of years people everywhere have asked questions that relate to space and time. This is a great journey of inquiry. *The Bible* begins with the story of creation that includes a description of how space and time came into being. Societies in many parts of the world provided a variety of religious, philosophical, as well as scientific explanations of the nature that surrounded them.

When Isaac Newton encoded the laws of mechanics in the 17th century in the language of mathematics, there seemed to be a clear vision of what space and time meant. But the mystery of space and especially time thickened in the beginning of the 20th century with Einstein's discovery of relativity.

In the 21st century M-theory advocates a unified theory of matter and forces based on 10- space and 1-time dimensions. Two-time physics envisions a greater unification in a space–time with one extra space and one extra time dimensions beyond those of M-theory. With its extra 1+1 dimensions, 2T-physics reveals hidden relationships among physical systems not captured by the ordinary formulation of physics in ordinary space–time at all scales of distance or of energy.

The universe around us is not as simple as it seems to be at first sight. Let's begin by imagining how a little boy, such as my grandson Isaac in Fig. 1.1, is likely to interpret his world if left on his own without guidance. Everyday he sees that it becomes light when the Sun rises and then it becomes dark when the Sun sets. The Sun, the Moon, and all the stars seem to be going around the Earth. Everybody would agree with Isaac on these observations that all of us can verify every single day with our own eyes. Armed with this assurance, soon he is likely to come to the conclusion, as the early man did, that the Earth is the center of the universe.

It seems so logical, but yet this is not true at all, as we know well today. What is missed at first sight is the rotation of the Earth around the south–north pole axis, which is sufficiently slow so as not to be felt by Earth's residents. For Earth's

I. Bars, J. Terning, Founding Editor: F. Nekoogar, *Extra Dimensions in Space and Time*, Multiversal Journeys, DOI 10.1007/978-0-387-77638-5_1,

Fig. 1.1 Isaac beginning to discover nature (Credits, Julie Culver.)

inhabitants, this self-rotation creates the illusion that the universe rotates around the Earth.

However, if Isaac got on a spaceship and landed on the Moon that also rotates around its own axis, the Moon would then seem to be the center of the universe! Of course, we can tell from our perspective that it is not so without any hesitation. In fact, Isaac would be able to see clearly from his spaceship that both the Earth and the Moon do indeed rotate around themselves. The early man's initial conclusion that seemed so solid is easily shattered by today's routine experiment of space travel. The observations of the Earth's residents were not wrong, but the interpretation was, so a more subtle theory of the universe had to be developed.

Nowadays in the space age we describe all of the above in a few lines, but historically it was extremely difficult to reach the correct conclusion. That the Earth is the center of the universe became the dominant view of all humanity for several thousands of years. This was reinforced by religious beliefs as well as by elaborate theories advocated by philosophers, astrologers, and astronomers. Observing the heavens and the stars was a serious undertaking in many societies, including the ancient Babylonian, Egyptian, Jewish, Greek, Arab, Aztec, and many European civilizations. These were done for both divination or religious guidance through astrology and sheer curiosity that eventually led to philosophical and scientific studies.

A mathematical theory of the heavens, with the Earth at the center of the universe, was developed by the Greek astronomer Ptolemy (AD 85–165) in the Roman province of Egypt. Astronomers at that time were puzzled by the motion of some bright objects in the night sky which they called "planets," meaning *wandering stars*. Planets were distinguished because they moved across the sky in relation to the other stars. The motion of some planets, which at times seemed to change their smooth trajectories and begin to move backward for a while and then again forward, was baffling. In the *Almagest*, Ptolemy developed a complex mathematical scheme based on geometry that accounted for much of the astronomical movements of the seven planets known at that time. In this theory, a large rapidly rotating outer sphere contained the stars, while planets were embedded in smaller rotating spheres, one for each planet. The theory was especially convincing because a precise mathematical description of planetary motion around a static earth apparently fitted well the observations. Ptolemy's theory of the universe, represented in the model of Fig. 1.2 with the Earth at the center, became the dominant view in the Western world for 13 long centuries.

Fig. 1.2 Ptolemy's universe (Credits, INMA.)

However, additional planets were found eventually and those were harder to describe with Ptolemy's spheres. The scheme had to be revised and became too elaborate and less and less convincing. Finally the Polish astronomer Nicolaus Copernicus (1473–1543) proposed the *heliocentric* theory, that the Sun, and not the Earth, was the center of the universe, as seen in Fig. 1.3. The Earth was not static but, like the other planets, it moved uniformly around the Sun. This revolutionary approach could account more simply for the strange planetary motion as observed from the Earth, because the planets did not all move at the same rate. Copernicus described the planetary orbits as a superposition of circles.

Galileo (1564–1642), an Italian physicist, mathematician, and astronomer, supported Copernicus' view. He was one of the originators of the scientific method, which required quantitative experiments whose results could be analyzed with mathematical precision. He believed that the laws of nature must have a mathematical description. He made important contributions to the understanding of the laws of motion under the influence of gravity and other effective forces. He also built a small telescope and with it he discovered that the planet Jupiter had four moons circling it. This shattered Ptolemy's universe, since by this verifiable observation the Earth could not be the center for Jupiter's moons. It was no longer a matter of just theory, the correct model was experimentally evident! However, many astronomers and philosophers who believed firmly in Ptolemy's model, initially refused to believe

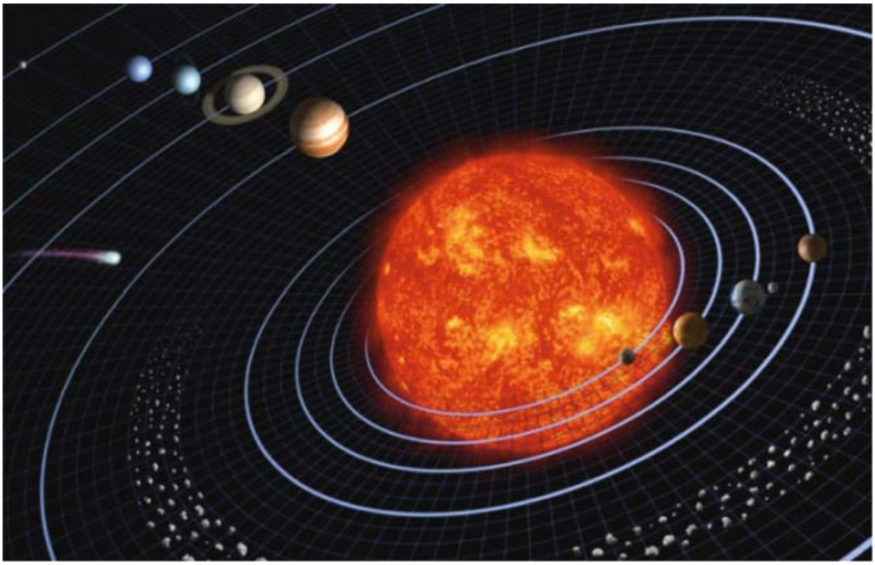

Fig. 1.3 Copernicus' universe. (Credits, NASA.)

that Galileo could have discovered such a thing. Galileo was accused of being a heretic by the Catholic Church, because the church, influenced by the literal interpretation of the Scriptures, also held the view that the Earth had to be the center of the universe. Galileo was put under house arrest by the Inquisition. To be able to save his life he had to recant, and he died in disgrace, condemned by the church.

The Copernican Revolution created a new perspective which led eventually to the German astronomer and mathematician Kepler (1571–1630) to develop the laws of planetary motion. Kepler had a much simpler mathematical description of each planetary orbit as being a well-defined single ellipse with the Sun sitting at one of its foci. Kepler's laws provided one of the foundations for Isaac Newton's theory of *universal gravitation*. By universal gravity, Newton meant that the gravitational force experienced on the Earth (like a falling apple) has the same origin as the gravitational force that holds the planets around the Sun. Newton was able to derive mathematically the elliptical planetary orbits from his laws of mechanics that followed from the simple statement

$$Force = Mass \times Acceleration,$$

where the force was due to the gravitational attraction of a relatively small planet to a large central Sun.

Except for small corrections that were explained later by Einstein's theory of general relativity, Newton's laws basically explain the motion of all objects in the solar system.

There are a lot more phenomena in the universe beyond the solar system, how are they described? By the end of the 19th century, physicists and astronomers had become convinced that it was possible to understand everything in the universe by using only Newton's laws of mechanics, his theory of universal gravity, and Maxwell's laws of electricity and magnetism.

However this order began to change right at the beginning of the 20th century with the dual discoveries by Planck and Einstein. In 1902 Planck proposed the quantum of energy which later led to quantum mechanics as the correct way to describe motion for subatomic distance scales. In 1905 Einstein proposed special relativity for the correct description of fast moving objects and in 1916 he introduced general relativity for the correct description of the gravitational force. General relativity includes extreme situations, when the gravitational force becomes very strong, such as the inside of black holes, or the time of the Big Bang which marks the birth of our universe.

Guided by these new insights, our horizons of the size and scope of the universe changed dramatically during the 20th century. This came about through further experimental discoveries made with large telescopes that explored remote parts of space far beyond our own galaxy, and with powerful accelerators that looked deep into the structure of matter in the subatomic and subnuclear regimes.

Hubble's observations of galaxies in the 1920s made it evident that the solar system is only a tiny part of the Milky Way galaxy. A galaxy is a conglomeration of billions of stars of various sizes similar to the Sun, which are held together as a dynamical system by gravitational attraction. Today it is estimated that there exists at least hundreds of billions of, maybe several trillion, galaxies in the universe.

It was at first thought that the matter contained in each galaxy is the same basic matter we see on the Earth, namely matter made up of quarks and leptons and the stuff that holds them together as explained in the next section. Not only that! According to later discoveries which refined this picture, there is more than meets the eye. As shown in Fig. 1.4, within each galaxy there is some additional matter that does not shine, which is called dark matter. The familiar matter has been estimated to make up only about 3–5% of the total energy of the universe, while dark matter accounts for 22–25% of it. The remaining almost 70–75% of the total energy is called dark energy, which is different than both ordinary and dark matter.

Even though we are aware of their existence, the nature of both dark matter and dark energy is poorly understood today. Dark matter responds to the gravitational force like ordinary matter does, but it does not seem to respond to the other familiar strong, weak, and electromagnetic forces. A hypothesis called WIMP, which stands for weakly interacting massive particles, attempts to partially describe dark matter. On the other hand, dark energy does not even behave like matter particles at all, but it still responds to the gravitational force although in a somewhat different way.

Starting with Hubble in the 1920s, it has been established that the universe as a whole is in motion and in a state of expansion. This is not understood in terms of Newton's laws. It is Einstein's general relativity that comes into play to explain it. Everything we observe today was all created in a Big Bang which initiated the expansion of the universe.